Transition to EMU

European Monetary System:1979-1998

Lasted 20 years (serious speculative attack in 1992)

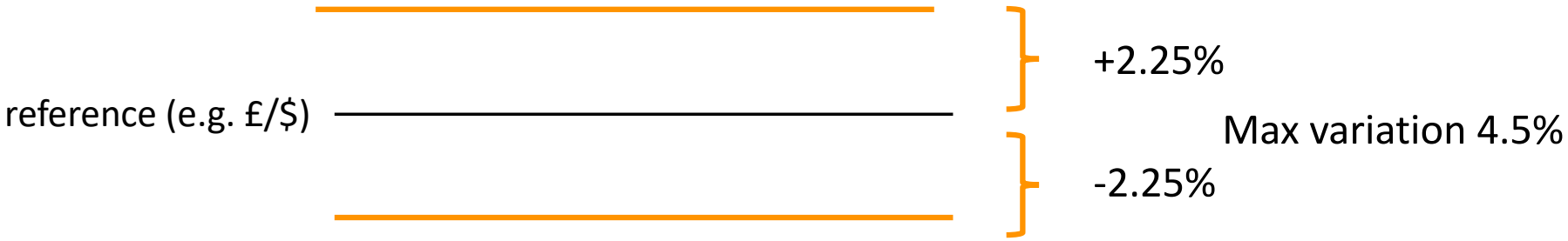
One of the longest examples, together with Bretton Woods, of exchange rate agreements

The outset

August 1971 Nixon: dollar inconvertibility (parity since 1944 35$ ounce);

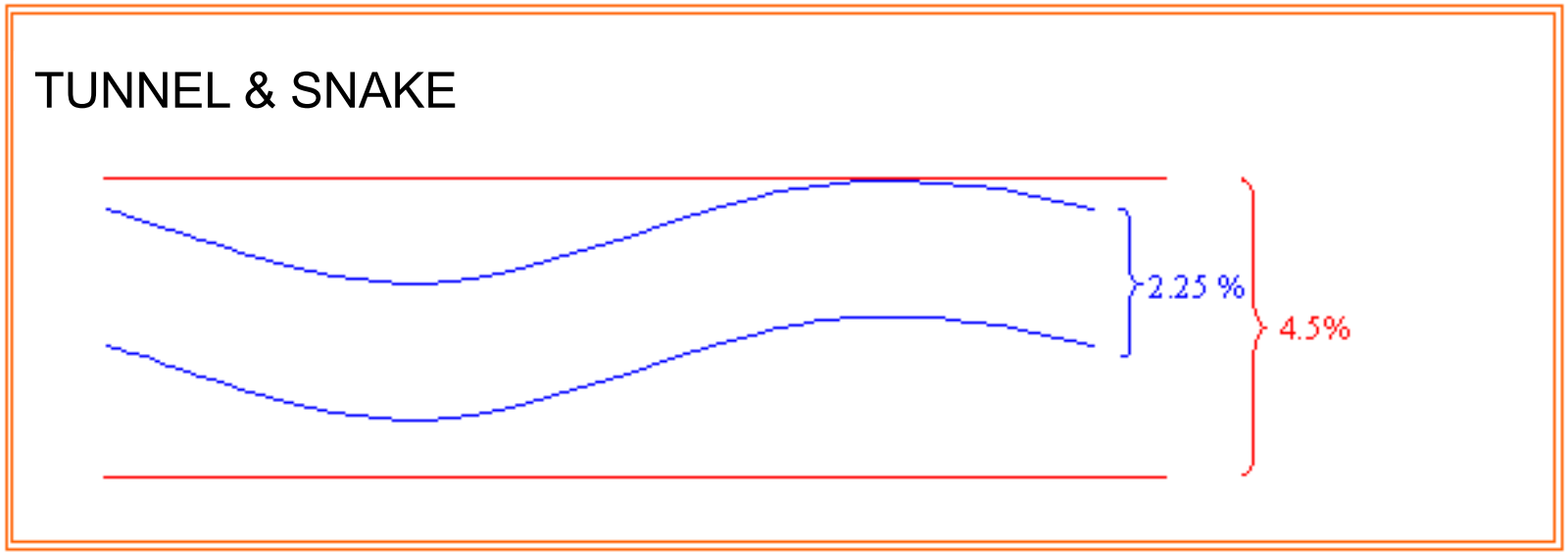
The dollar reference was lost

December 1971 G10: Smithsonian Agreement (dollar devalues 8% and new flexibility range +/-2.25% wrt dollar for other currencies)



The Tunnel

1972: 6 countries of the European Economic Community (EEC) introduce the ‘Snake’.
The range of variation was restricted to +-1.125% for currency within EEC



The reason for the snake

Since margins wrt to the dollar were +/-2.25%, margins wrt other currencies were +-4.5%.

Example:

		a	**b**		**a/b**	
		Lira/$	DM/$		Lira/DM	
		1500	2.5		600	
1	2.25%	1533.75	2.55625		600	
2	-2.25%	1466.25	2.44375		600	
				a1/b2	627.621483	
				a2/b1	573.594132	
				max var	0.09419091	

From Snake to the European Monetary System (EMS)

Snake: UK, IR and ITA left it in 1972, FR in1974

Problems: difficulties to remain within the range of fluctuations due to different inflation rate. Only the German area (Ger, Benelux, Denmark) remained in the snake

In the 1970's various capital controls were introduced

EMS Objectives:

1. Extend the 'Snake' area of coordination but with **greater equity among currencies**.
2. Provide mechanisms of credit for weak currencies

For FR and ITA, EMS being part of EMS was a relevant part of the fight against inflation. But after a few years, it became clear that the EMS marked the acceptance of Germany macroeconomic priorities (See next slide)

Notice also: all these countries (FR, IT e GER) shared the common objective of Europe unification and of a European identity vis à vis USA.

EMS and its novelties

ECU = European Currency Unit: a currency basket based on a weighted average of EMS currencies. ECU was a 'notional' currency.

ECU: for each currency i:

$$ECU_i = \sum_{j=1,n} a_j S_{ij}$$

a_j is the weight of currency j in the basket based on GDP and trade shares e S_{ij} is the bilateral exchange rate of currency j and currency i.

Note: if S are the central parity rates, then the value of ECU is the central value

If S are market rates, then the ECU value is the market value

The difference between the two is an important indicator of deviation of the currency wrt the average

EMS and its novelties

ECU = European Currency Unit: a currency basket based on a weighted average of EMS currencies. ECU was a 'notional' currency.

ERM = Exchange Rate Mechanism defined the central parity of each currency to the ECU

Bilateral parities network defined by central parities

Example: £/ECU=a and DM/ECU=b → £/DM=a/b.

Fluctuation bands

Narrow band - Standard:+/-2.25%

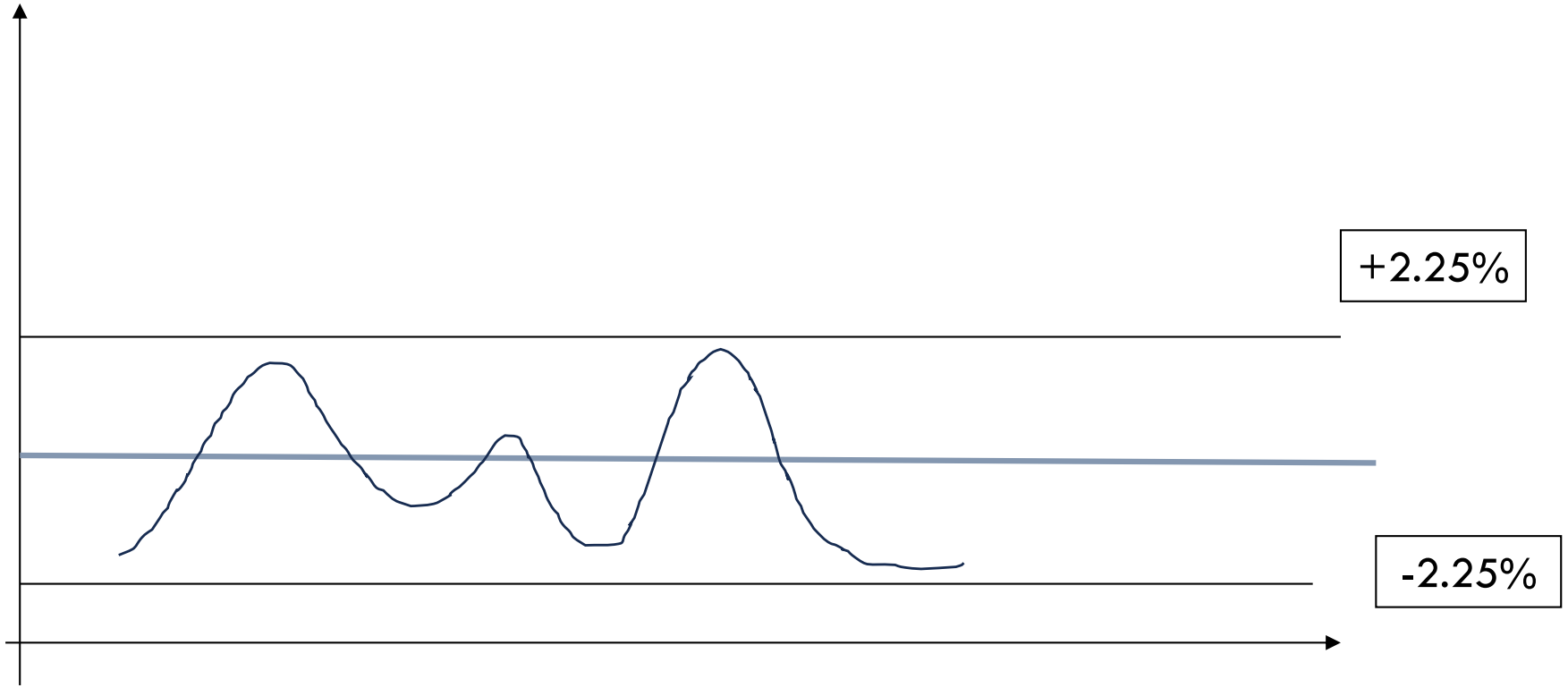
Wide band (allowed to Italy, Spain, UK): +/- 6%

From August 1993 : +/- 15% for all currencies.

Monetary policy intervention: Automatic and compulsory interventions

When a currency approached the upper or lower limit of its band, **the central banks of both currencies were obliged to intervene** by buying or selling its own currency to defend the parity.

Compulsory interventions at the upper and lower limit



EMS and its novelties

Other support mechanisms

- Very Short-Term Financing: the main automatic credit mechanism

The central bank defending its currency could obtain very short-term credit from the central bank of the other currency. These credits were designed to finance exchange-market interventions and had to be repaid quickly — normally within 45 days.

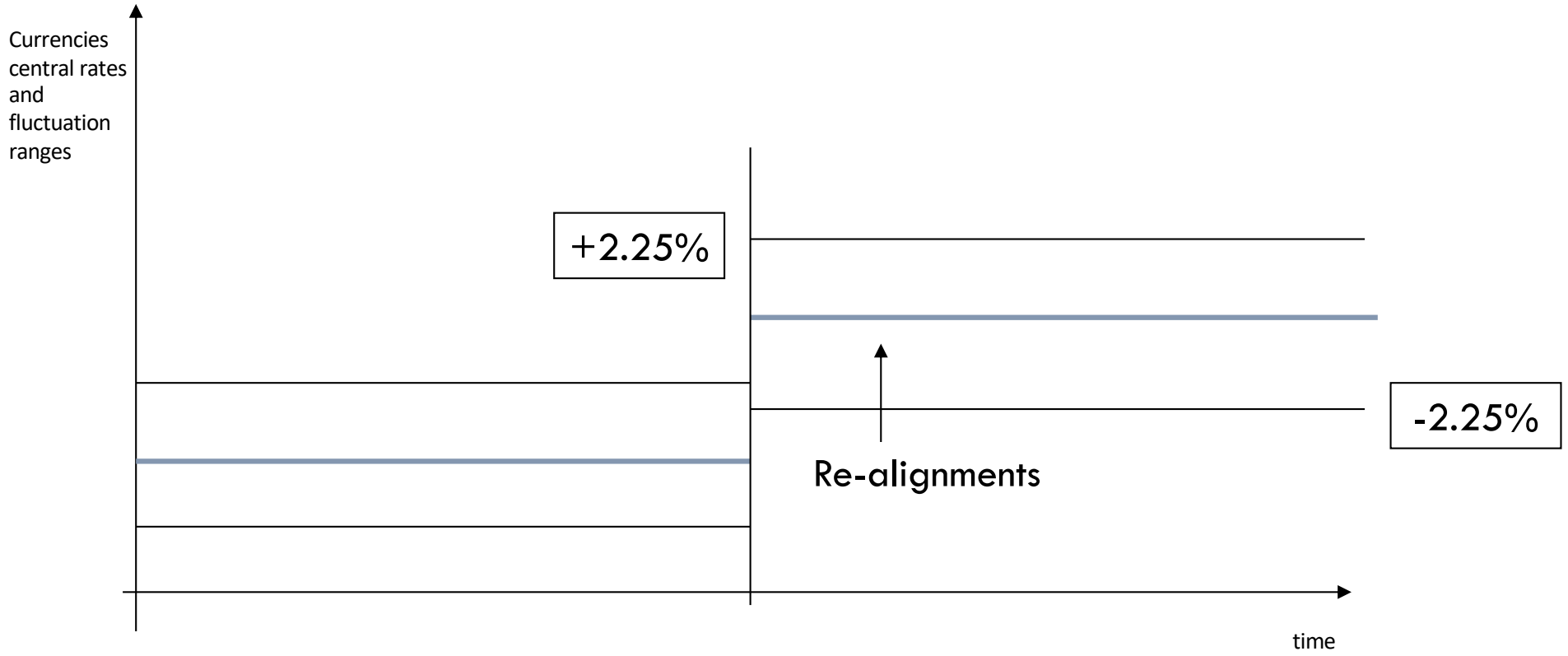
- Short- and medium-term credit facilities

Realignments

Central parities could be changed at unanimity: REALIGNMENTS

This was a radical new departure because exchange rates could only be changed by **mutual agreement between participating Member States** and the European Commission – an unprecedented pooling of monetary sovereignty.

Realignments



Four phases of ERM

1. **1979-83** Frequent realignments as policies and inflation rates were still divergent
2. **1983-87** Less realignments as inflation rates start converging
3. **1987-92** No realignments. Italian lira entered the narrow band; Spain, followed by UK and Portugal join between 1989 and 1990, with a wide band. Final capital controls were removed; Maastricht Treaty was signed; Germany reunification.
   1992 Speculations against bilateral exchange rates of a few over-valued currencies. Italian lira and Pound sterling leave ERM in Sept 1992.
4. **1993-1998** bands are widened to +/-15% . Italian lira enters again in 1996; large devaluation between 1992 and 1995.

The nth country in an exchange rate agreement

Note: Suppose there are *n* countries in an exchange agreement.

In order to keep the exchange rates fixed, only ***n-1*** central banks will have to devote their monetary policy to the exchange rate objective. The nth central bank can use the monetary policy to target its own objective.

If there are ***n*** countries, the exchange rates to be set are $(n^2-n)/2$

Example: 3 countries → 3 exchange rates: Lira/Franc, Franc/DM, Lira/DM.

BoI→ Lira/FR; Bank of France→ Fr/DM;

→Lira/DM is automatically fixed, as it is given by Lira/Fr x Fr/DM

The Bundesbank need not use the monetary policy to keep the exchange rate at the agreed level → it can choose the monetary policy 'regardless' of the exchange rate commitment.

Symmetric and asymmetric solution

Which country is going to be this n^{th} one?

Two possible solutions to this indeterminacy:

- the symmetrical or cooperative solution
- the asymmetrical or hegemonic solution.

In the EMS, the solution was hegemonic and the nth country was Germany

The nth country

The choice can be traced back to two interpretations:

1. the leading country is the 'monetarily' strongest one, i.e. the one that can maintain its monetary policy without changing the exchange rate
2. the leading country is the most credible one: in the long run it is 'Pareto efficient' to adapt to a situation of low inflation rather than to adapt to a situation of high inflation

Pros and cons of the hegemonic solution

Advantages

The leading country acts as an anchor and help regulate countries with inflationary tendencies.

Disadvantages

In the presence of recessionary cycles in peripheral countries, they must adopt pro-cyclical, destabilizing policies.

Why?

The leading country sterilizes incoming liquidity flows, which occur when the central banks of other countries intervene to support their currency in crisis situations.

As a result, the adjustments necessary to support the exchange rate at the decided level (in particular the increase in the interest rate if the central bank operates to support the exchange rate from depreciation pressures) are all borne by the peripheral countries.

1992 EMS crisis

German reunification:

increase public spending, increase deficit, inflationary pressure
→Bundesbank increases interest rates.
Meanwhile, in the rest of Europe, the severe recession underway required a reduction in interest rates. But they could not be lowered unless they were realigned and devalued

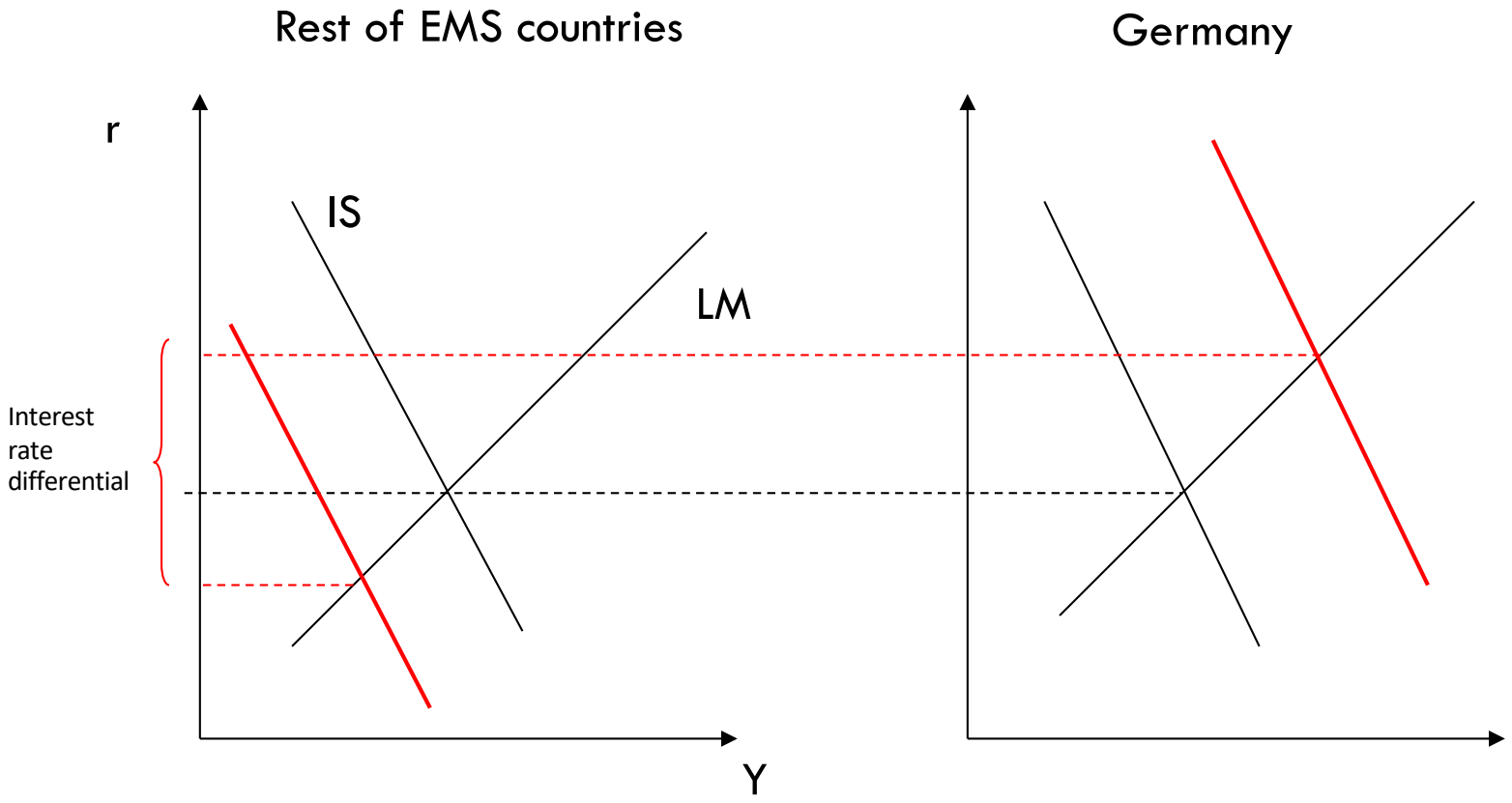
For political and credibility reasons, and also because in some cases the fundamentals did not require it, governments decided not to realign. But high interest rates in a recession led to expectations of realignment by the markets.

In the end, the attack was such that it led to the crisis of the ERM.
The economic policy conflict between Germany and the rest of the EC became evident

1992 EMS crisis

German reunification increases public spending. Bundesbank is not accommodative → interest rate increase.

Meanwhile, in the rest of Europe → severe recession and reduction in interest rates.

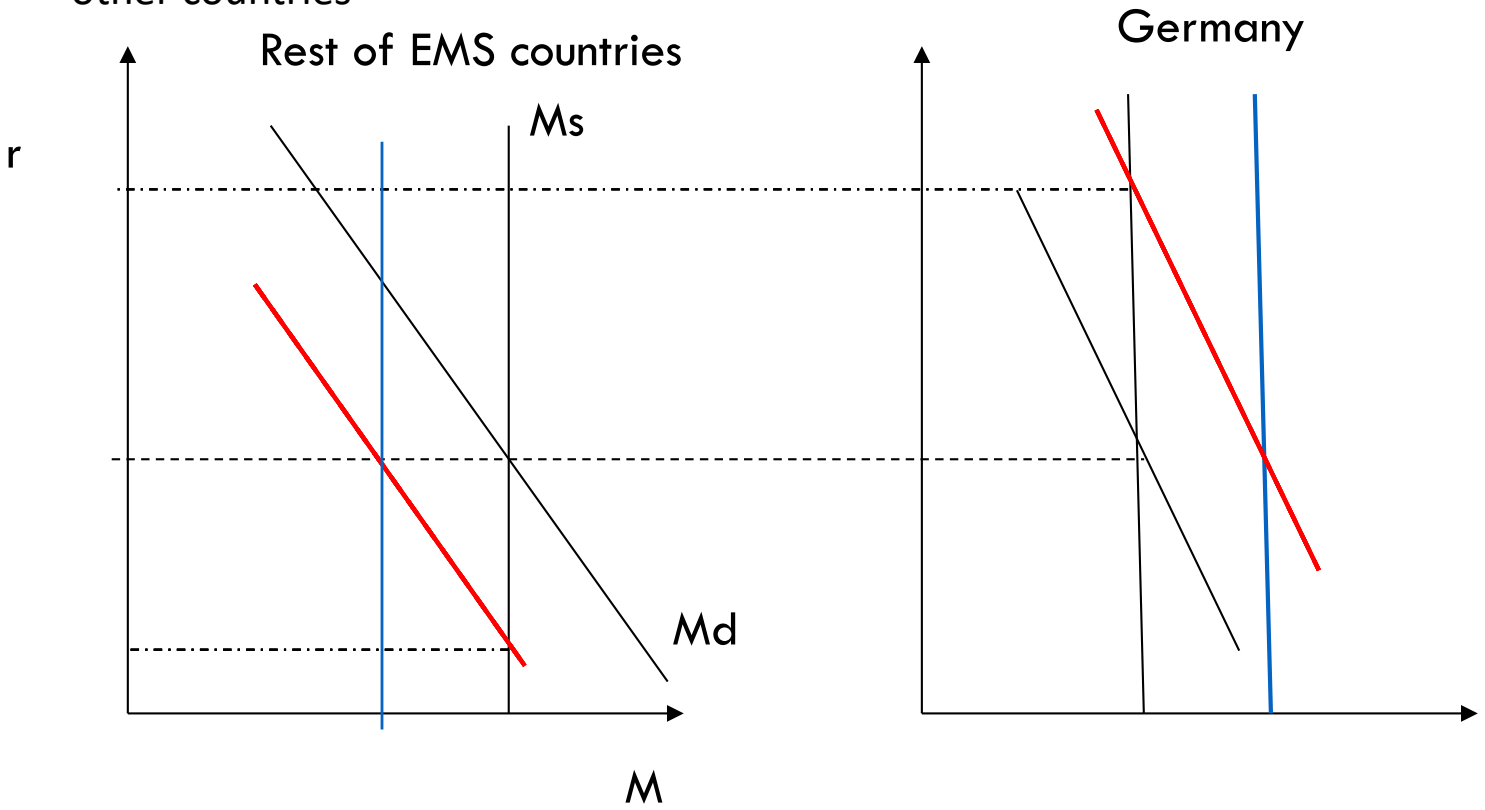


Monetary policy

In the money market, demand varies in the same direction as IS, while the change in supply depends on the choice of monetary authorities.

Hypothesis 1: No realignment (exchange rate remains fixed)

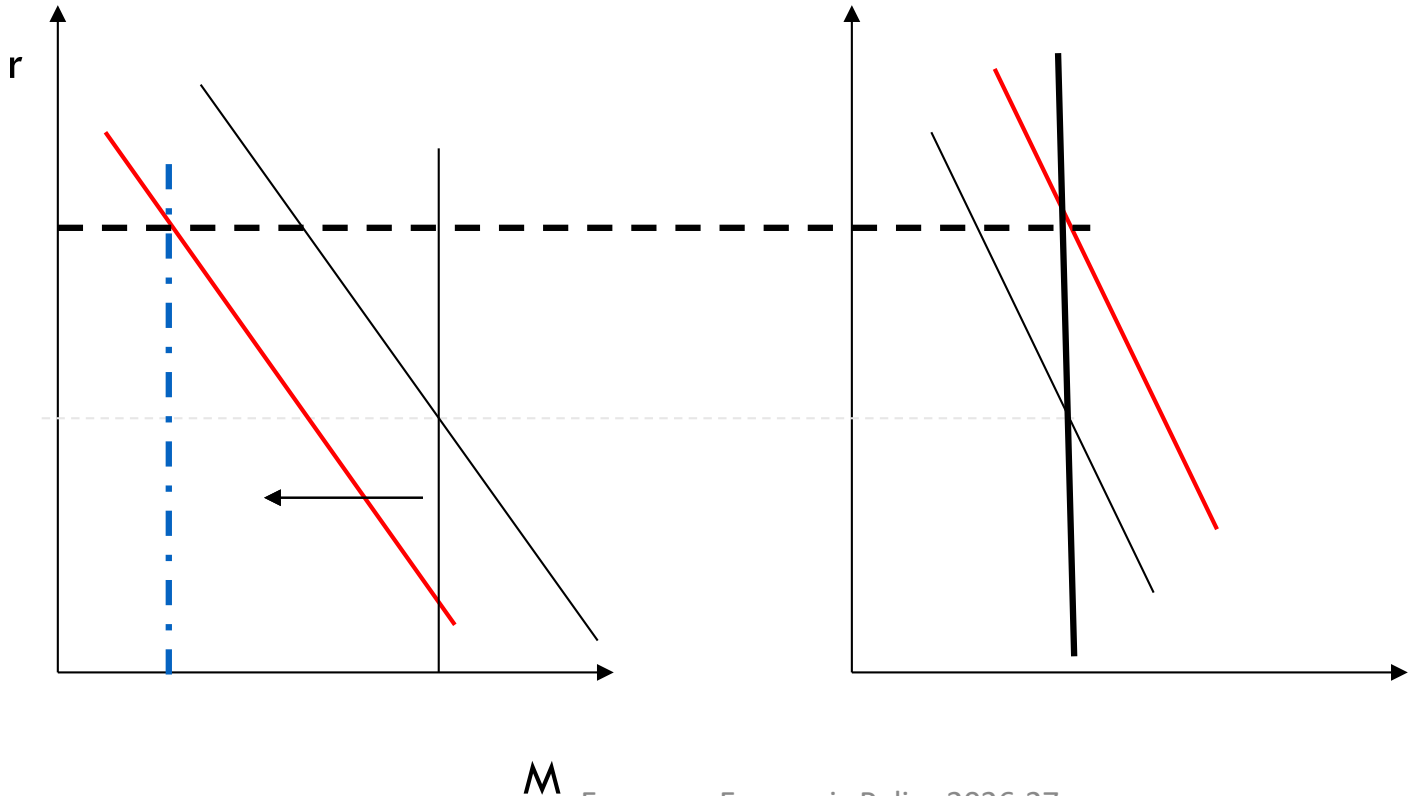
Symmetric solution: rising interest rate in Ger, capital inflows, improving the BoP; with a fixed exchange rate, a surplus in the BoP is generated, which translates into an increase in the money supply (blue line) and a consequent reduction in the interest rate (and probable increase in inflation). The increase in money supply in Ger is mirrored by a decline in the other countries



Monetary policy

Hypothesis 1: No realignment

Asymmetric solution: increase in the interest rate in Ger, capital inflow, improvement of the BoP which would automatically lead to an increase in the money supply (as in the case seen above) BUT the German authorities decide to STERILIZE this change in money; therefore the money supply remains unchanged. To maintain the fixed exchange rate, the adjustment is therefore borne entirely by the other countries that have to reduce their money supply by more than in the case seen before, i.e. they need to adopt a restrictive monetary policy (restrictive policies, in a period of recession are pro-cyclical policies!)



Monetary policy

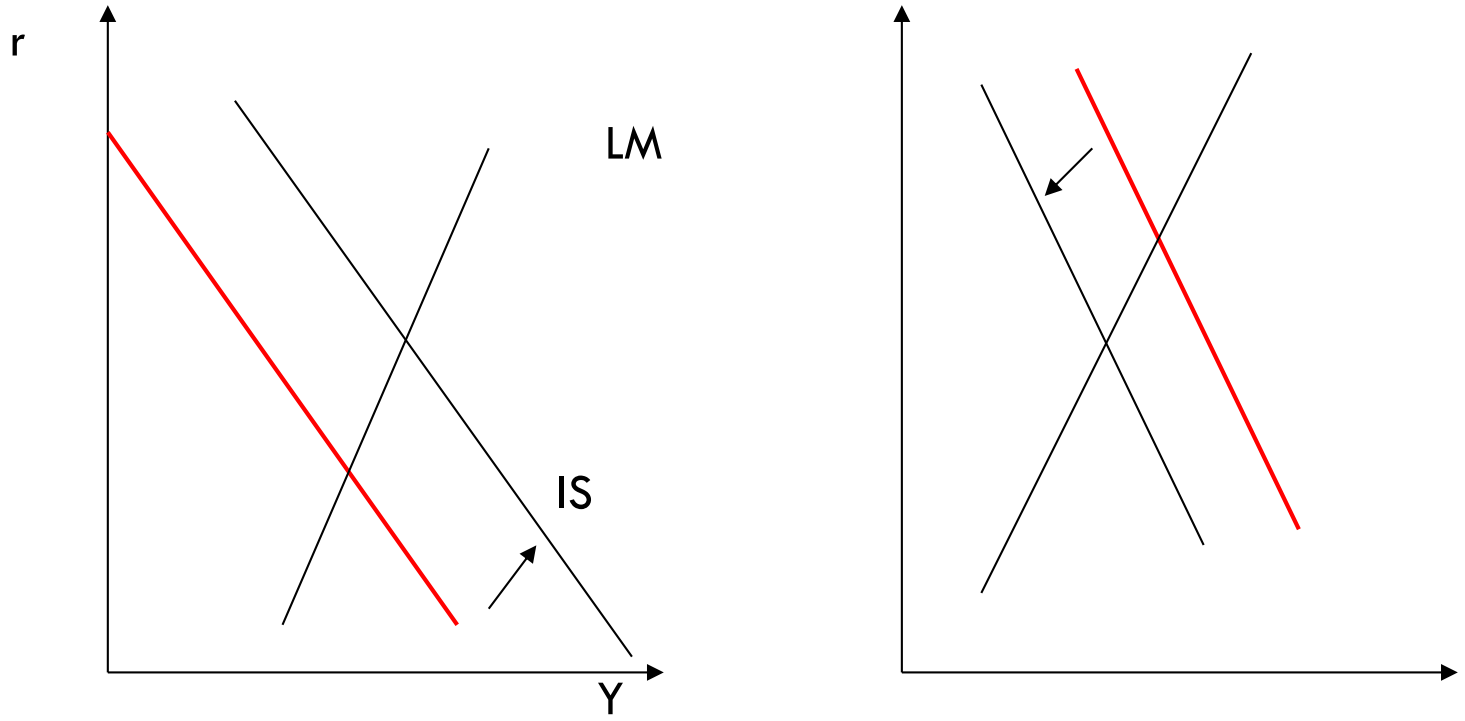
Note: in the case of a fixed exchange rate and a symmetric solution, competitiveness will be restored, over time, by price changes. The increase in aggregate demand and money pushes for an increase prices in Germany as recession pushes prices to fall in other countries.

2nd Hypothesis: Realignment of central rates

The DM revalues against all other currencies.

Competitiveness is restored by the change in the nominal exchange rate.

The equilibrium is re-established through the effects on aggregate demand of the devaluation/revaluation



From the 1992 crisis to the broad bands

For political and credibility reasons and also because in some cases the fundamentals did not require it, the governments (France and the UK) did not agree on the possible realignment, supported instead by Ger and Italy.

- Interest rates that were too high for a period of recession led to expectations of realignment on the part of the markets.
- This was also supported by the Danish NO to the Maastricht Treaty, which cast doubt on the project of monetary union.
- The markets, anticipating a devaluation, attacked the weak currencies. The authorities failed to avoid this attack (unlimited credit was not given to central banks)
- In September 1992, the UK and Italy decided not to intervene to keep the currencies within the band granted by the AECs and the EMS entered into crisis.

From the 1992 crisis to the broadbands

Here is a summary of the devaluations that took place between May and July 1993

Spanish peseta: The peseta depreciated twice in this period, with a cumulative devaluation of around 15-20%.

Portuguese escudo: The escudo has come under similar pressure and has been devalued by around 6-10%.

Irish pound: The Irish currency has depreciated by about 10%.

French franc and Belgian franc: Both currencies came under pressure, but held exchange rates due to strong central bank interventions.

Danish krone: Denmark has maintained the parity of the krona within the ERM through interventions, avoiding devaluation.

So, the countries decided to widen the bands to +-15%.

The economic policy conflict between Germany and the rest of the EC became apparent.

Impossible trio: Independent monetary policies, fixed exchange rates and capital mobility.

The Maastricht Treaty

- The Maastricht Treaty was signed in 1991.
- It is the blueprint for progress towards monetary unification in Europe.
- It is based on two principles:
  - Gradualism: the transition towards monetary union in Europe is seen as a gradual one
  - Convergence criteria: entry into the union is made conditional on satisfying convergence criteria.

Convergence criteria

For each candidate country:

(1) inflation rate $\leq$ average of three lowest inflation rates in the group of candidate countries + 1.5pp;

(2) long-term interest rate $\leq$ average observed in the three low-inflation countries + 2pp;

(3) joined the exchange rate mechanism of the EMS and did not experience a devaluation during the two years preceding the entrance into EMU;

(4) government budget deficit $\leq$ 3% of its GDP

if this condition is not satisfied:

budget deficit should be declining continuously and substantially and come close to the 3%,

or the deviation from the reference value (3%) 'should be exceptional and temporary and remain close to the reference value', art. 104c(a));

(5) government debt $\leq$ 60%of GDP

if this condition is not satisfied:

government debt should 'diminish sufficiently and approach the reference value (60%) at a satisfactory pace', art. 104c(b)).

Why convergence requirements?

- The OCA theory stresses *micro*-economic conditions for a successful monetary union:
  - symmetry of shocks
  - wage and price flexibility
  - labour mobility
- The Treaty stresses *macro*-economic convergence but only related to monetary policy and public finance
  - inflation
  - interest rates
  - budgetary policies

Check your learning

- Which features of the EMS were crucial for its long survival?
- What was ECU?
- Why were European countries interested in an European exchange rate agreement?
- Why can speculative attacks on currencies happen only when exchange rates are fixed or somewhat constrained ?
- Would you say that asymmetric shocks were desrupting the economic stability of European countries even before the euro? Why?